

3.7 Implicit Differentiation

Most of the functions we have dealt with so far have been described by an equation of the form $y = f(x)$ that expresses y explicitly in terms of the variable x . We have learned rules for differentiating functions defined in this way. For example,

$$\begin{aligned}y &= x^2 + x + 1 \Rightarrow y' = 2x + 1; \\y &= \sin x + x^3 \Rightarrow y' = \cos x + 3x^2.\end{aligned}$$

A different situation occurs when we encounter equations like

$$\begin{aligned}x^3 + y^3 - 9xy &= 0. \\x^2y^2 - 2xy + \cos y &= 0\end{aligned}$$

These equations define an *implicit* relation between the variables x and y , meaning that a value of x determines one or more values of y , even though we do not have a simple formula for the y -values. In some cases, we may be able to solve such an equation for y as an explicit function (or even several functions) of x . When we cannot put an equation $F(x, y) = 0$ in the form $y = f(x)$ to differentiate it in the usual way, we may still be able to find $\frac{dy}{dx}$ by *implicit differentiation*. This section describes the technique.

① Implicit Differentiation

◆ Implicit Differentiation

1) Differentiate both sides of the equation with respect to x , treating y as a differentiable function of x .

2) Collect the terms with $\frac{dy}{dx}$ on one side of the equation and solve for $\frac{dy}{dx}$.

Note: If $y = f(x)$, then

$$\begin{aligned}(y^2)' &= 2y y'; \\(y^3)' &= 3y^2 y'; \\(y^4)' &= 4y^3 y'; \\(y^5)' &= 5y^4 y' .\end{aligned}$$

Example 1 Find the slope of the circle $x^2 + y^2 = 25$ at the point $(3, -4)$.

Example 2 Find $\frac{dy}{dx}$ if $y^2 = x^2 + \sin xy$.

Example 3 Find y' if $x^2 + y^2 = 1$.

Example 4 Find $\frac{dy}{dx}$ when $x^2 + xy + y^2 = 1$.

Example 5 Find y' when $x^2y^3 - xy + 2y^2 = x$.

② Tangent and Normal lines

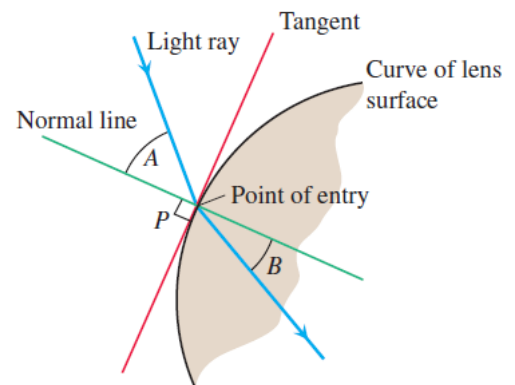
Recall: Two lines are orthogonal $\Rightarrow$ the product of their slopes is -1 .

Recall: The **normal** is the line perpendicular to the tangent of the curve at the point of entry.

For given curve $y = f(x)$, the tangent line and normal line to the curve at point (x_0, y_0) are

$$\text{Tangent line: } y - y_0 = f'(x_0)(x - x_0).$$

$$\text{Normal line: } y - y_0 = -\frac{1}{f'(x_0)}(x - x_0).$$



Example 6 Show that the point $(2,4)$ lies on the curve $x^3 + y^3 - 9xy = 0$. Then find the tangent and normal to the curve there.